RAJALAKSHMI ENGINEERING COLLEGE

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M2

Started on	Friday, 21 August 2026, 8:41 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 3:14 PM
Time taken	40 days 6 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>


int binarySearch(int arr[], int low, int high, int target) {
    if (low > high) {
        return -1;
    }

    int mid = low + (high - low) / 2;

    if (arr[mid] == target) {
        return mid;
    } else if (arr[mid] > target) {
        return binarySearch(arr, low, mid - 1, target);
    } else {
        return binarySearch(arr, mid + 1, high, target);
    }
}


int findPair(int arr[], int low, int high, int x, int *elem1, int *elem2) {
    for (int i = low; i <= high; i++) {
        int complement = x - arr[i];


        int j = binarySearch(arr, low, high, complement);


        if (j != -1 && j != i) {
            *elem1 = arr[i];
            *elem2 = arr[j];
            return 1;
        }
    }
    return 0;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int *arr = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int x;
    scanf("%d", &x);

    int elem1, elem2;
    if (findPair(arr, 0, n - 1, x, &elem1, &elem2)) {
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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